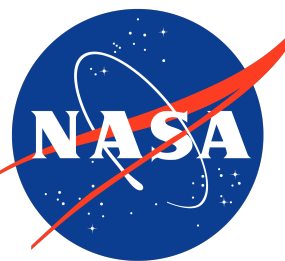


Unveiling the transferability of PLSR models for leaf trait estimation: lessons from a comprehensive analysis with a novel global dataset



Global Change Research Laboratory
For better understanding the complex interactions in the Earth system



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Motivation

- Leaf traits play an important role in modulating plant physiological processes and ecosystem biophysical and biochemical dynamics (Kattge et al., 2011).
- Quantifying and understanding the variations of leaf traits is essential for improving our prediction of the effects of environmental changes on terrestrial ecosystems.
- Partial least-squares regression (PLSR) models with leaf spectroscopy are widely applied for trait estimation, **but their transferability across space, time and plant functional types (PFTs) remains unclear.**

Questions:

- How well do the PLSR models transfer to new domains (new sites, PFTs and time)?
- What are the potential factors influencing the transferability of PLSR?

Methods

Compiled dataset:

- Paired leaf traits (Chla+b, Ccar, EWT, LMA) and spectra (450–2400 nm).
- 47,393 records, >700 species and eight PFTs at 90 globally-distributed locations across multiple seasons.
- 6,840 leaf samples for Chla+b, 4,233 samples for Ccar, 3,646 samples for EWT, and 45,417 leaf samples for LMA.

Methods:

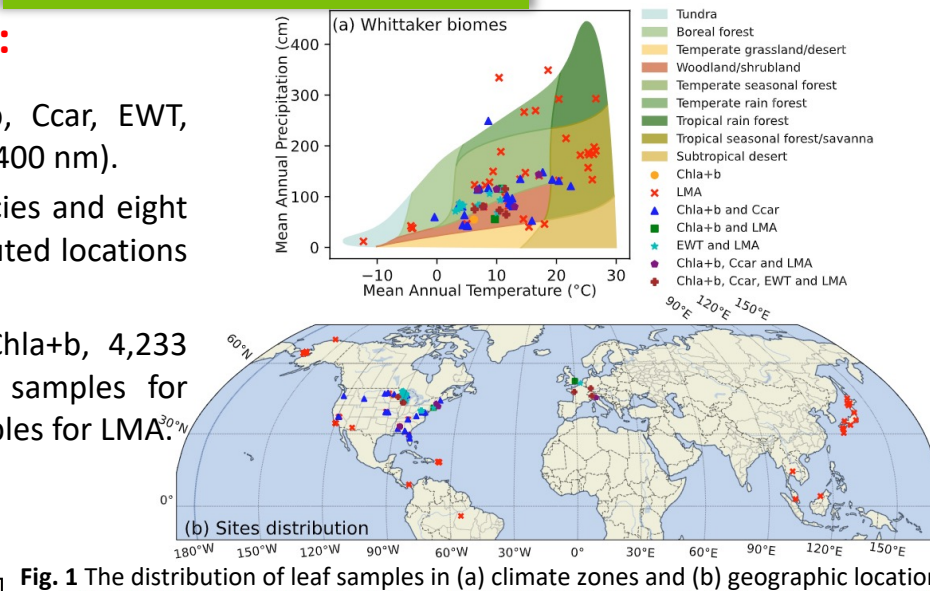
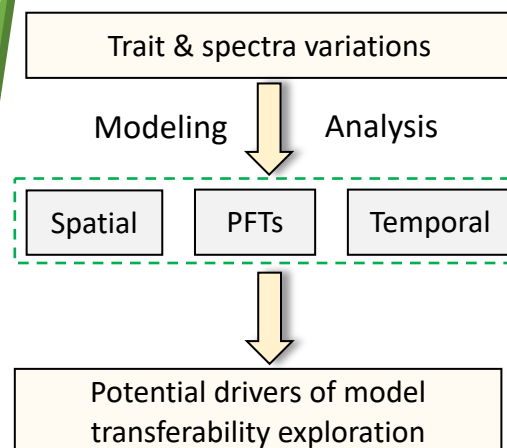


Fig. 1 The distribution of leaf samples in (a) climate zones and (b) geographic locations

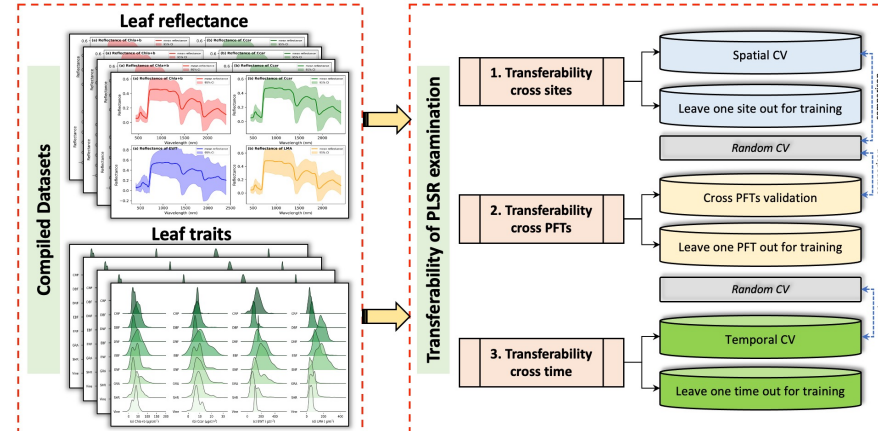


Fig.2 The framework of this study.

Results

Variations of leaf traits and leaf spectra

- Leaf traits and leaf spectra are highly variable across sites, PFTs and phenological stages.

Spatial, PFTs and temporal analysis

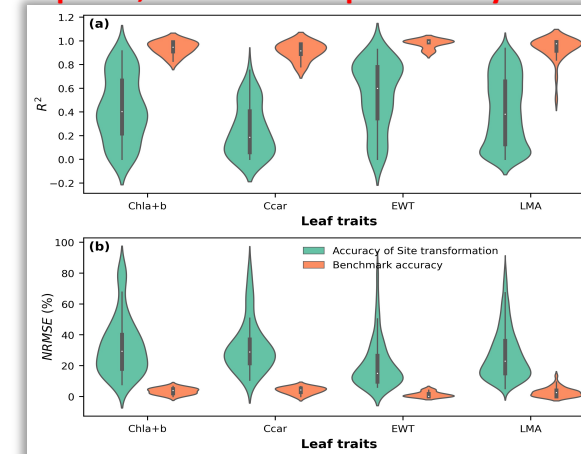


Fig.3 Cross-sites transformation.

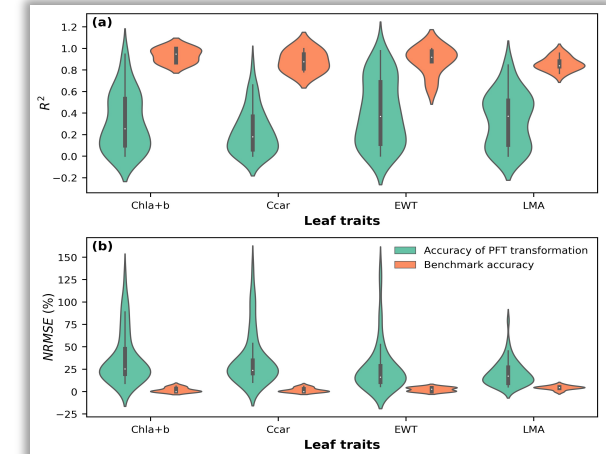


Fig.4 Cross-PFTs transformation.

- Predictive models can expect accurate trait predictions for new measurements falling within the range of the data used during model training.
- The model performance will **decline** when extrapolating to **new domains**.

Factors influencing the transferability of models

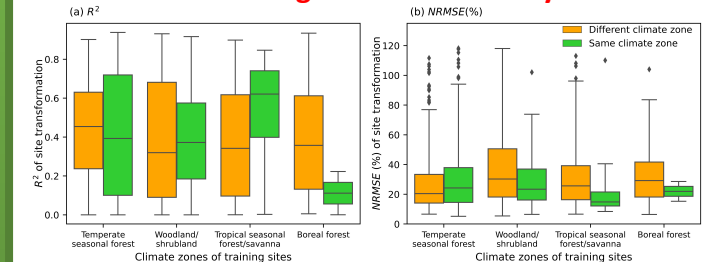


Fig. 6 Climate variables to explain the transferability of PLSR.

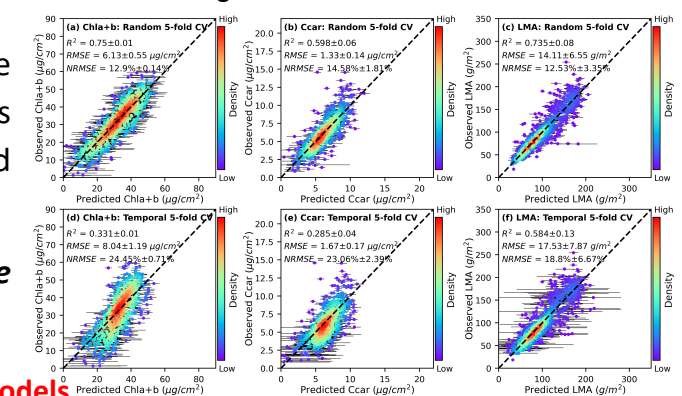


Fig.5 Cross-time transformation.

- No obvious influence of climate variables** on the transferability of PLSR models for trait prediction.
- Alpha and beta spectral diversity exhibited **positive relationships** in which greater similarity of a spectral diversity metric yielded greater PLSR transferability.

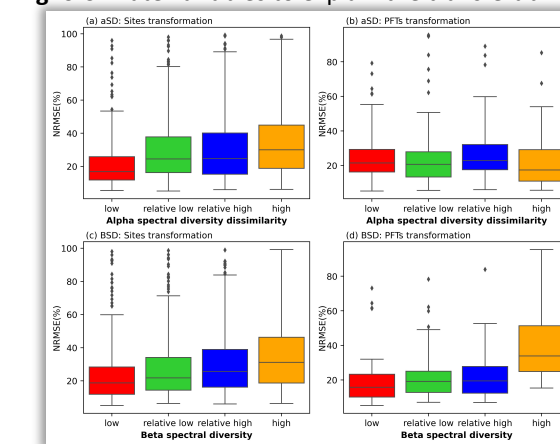


Fig. 7 Spectral diversity to explain the transferability of PLSR.

Take-home messages:

- Leaf spectroscopy and PLSR modeling are highly capable of quantifying leaf traits given sufficiently comprehensive training data.
- Model performance will significantly decline when extrapolating to novel contexts.
- Underscores the importance of incorporating greater spectral diversity in model training to boost its transferability.

Reference

[1] Kattge, J., Díaz, S., et al. 2011. TRY – a global database of plant traits. Glob. Chang. Biol. 17, 2905–2935.

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